Monte Carlo Path integral for the Harmonic Oscillator

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What is Path Integral?

What is Path Integral

- Formulation of Quantum Mechanics

1. Quantization

$$i\hbar\frac{\partial\psi}{\partial t} = -\frac{\hbar^2}{2m}\frac{\partial^2\psi}{\partial x^2} + V(x)\psi$$

- Schrödinger equation
- Physical quantities are operators

$$\hat{x} = x,\ \hat{p} = \frac{\hbar}{i}\frac{\partial}{\partial x},\ \hat{x}\hat{p} \neq \hat{p}\hat{x}$$

Heisenberg

Schrödinger

What is Path Integral

- Formulation of Quantum Mechanics

  2. Path Integral

$$K(x_f, x_i, t) = \langle x_f | e^{-i\hat{H}t/\hbar} | x_i \rangle = \int \mathcal{D}x \ e^{iS[x(t)]/\hbar}$$

$$S[x(t)] = \int dt \left(\frac{1}{2} m \dot{x}^2 - V(x) \right)$$

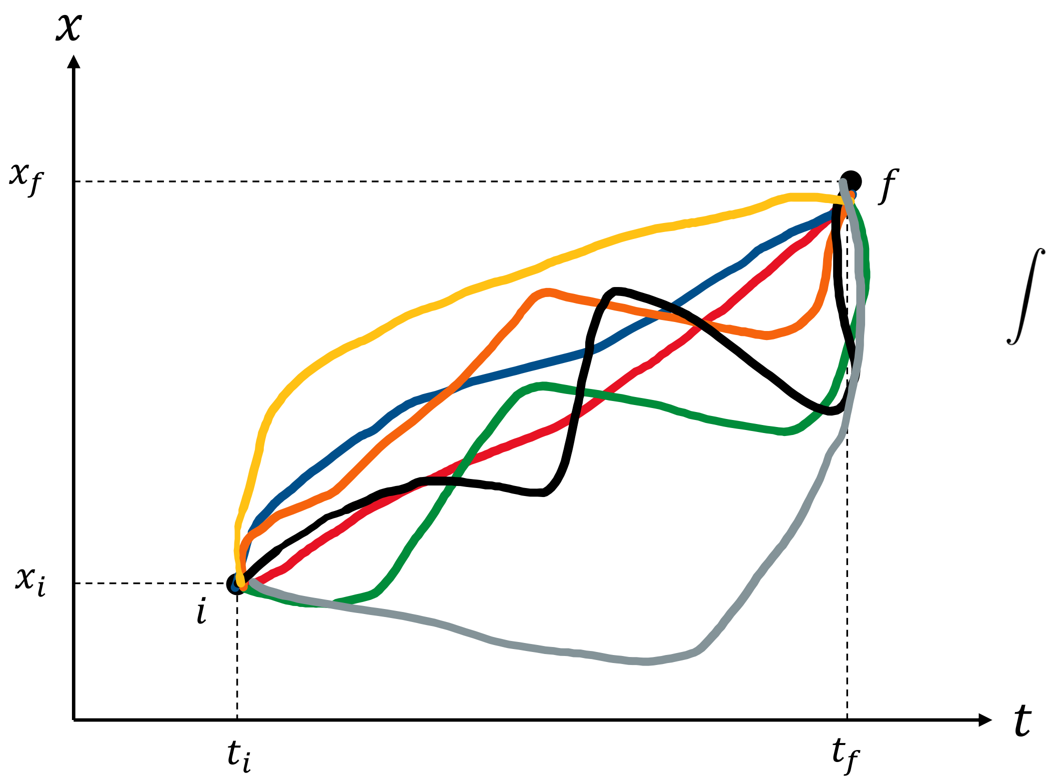
  - Propagator $K(x_f, x_i, t)$, action $S[x(t)]$
  - Physical quantities are variables (not operators)

$$xp = px$$

Feynman

What is Path Integral

$$K(x_f, x_i, t) = \langle x_f | e^{-i\hat{H}t/\hbar} | x_i \rangle = \int \mathcal{D}x \; e^{iS[x(t)]/\hbar}$$

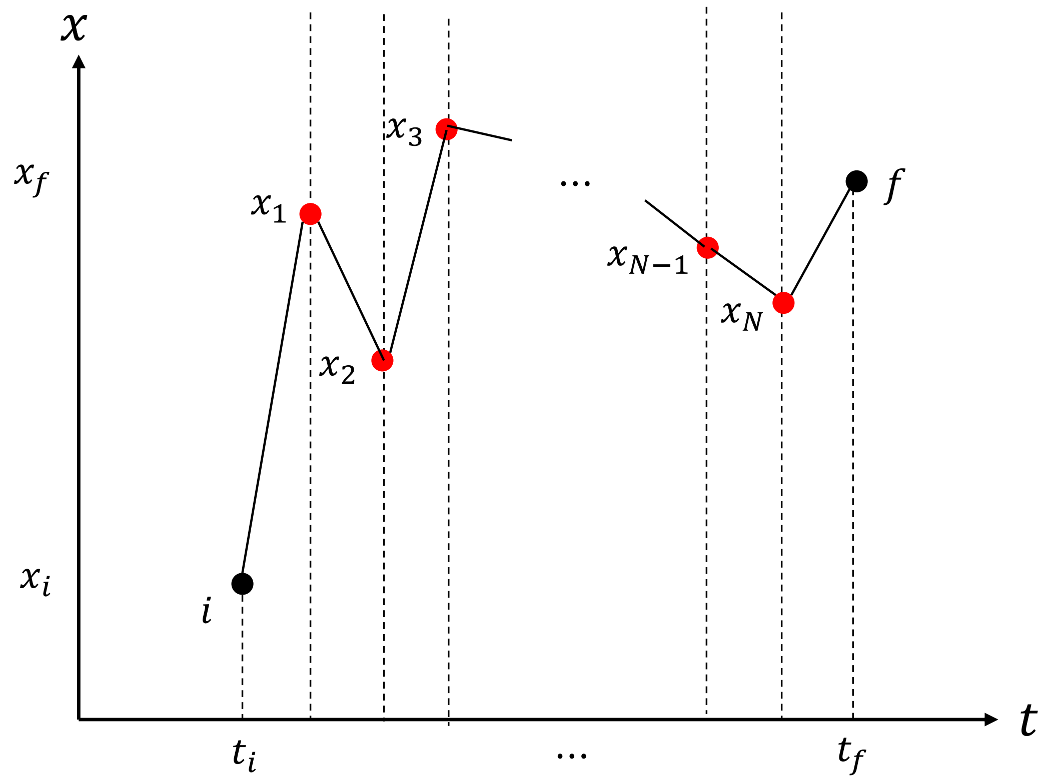


- Integrate for all possible paths with weight $e^{iS[x(t)]/\hbar}$

$$\int \mathcal{D}x \; e^{iS[x(t)]/\hbar} \approx \sum_{\text{all possible paths } x(t)} e^{iS[x(t)]/\hbar}$$

What is Path Integral

$$K(x_f, x_i, t) = \langle x_f | e^{-i\hat{H}t/\hbar} | x_i \rangle = \int \mathcal{D}x \ e^{iS[x(t)]/\hbar}$$



- Manipulation of x_j ($j = 1, \cdots N$) provides the variation of path
- N-dimensional integration

PIMC: Path Integral Monte Carlo

PIMC

- Wick rotation: $t = -i\tau$

$$\frac{i}{\hbar}S[x(t)] = \frac{i}{\hbar}\int -i d\tau \left(-\frac{1}{2}m\left(\frac{dx}{d\tau}\right)^2 - V(x)\right)$$
$$= \frac{-1}{\hbar}\int d\tau \left(\frac{1}{2}m\left(\frac{dx}{d\tau}\right)^2 + V(x)\right) = \frac{-1}{\hbar}S_E[x(\tau)]$$

$$K_E(x_f, x_i, \tau) = \langle x_f | e^{-\hat{H}\tau/\hbar} | x_i \rangle = \int \mathcal{D}x \; e^{-S_E[x(\tau)]/\hbar}$$

PIMC

- Partition function

$$Z = \mathrm{tr}(e^{-\beta \hat{H}}) = \int dx\ \langle x | e^{-\beta \hat{H}} | x \rangle, \quad \left(\beta = \frac{1}{k_B T} \right)$$

$$K_E(x_f, x_i, \tau) = \langle x_f | e^{-\hat{H}\tau/\hbar} | x_i \rangle = \int \mathcal{D}x\ e^{-S_E[x(\tau)]/\hbar}$$

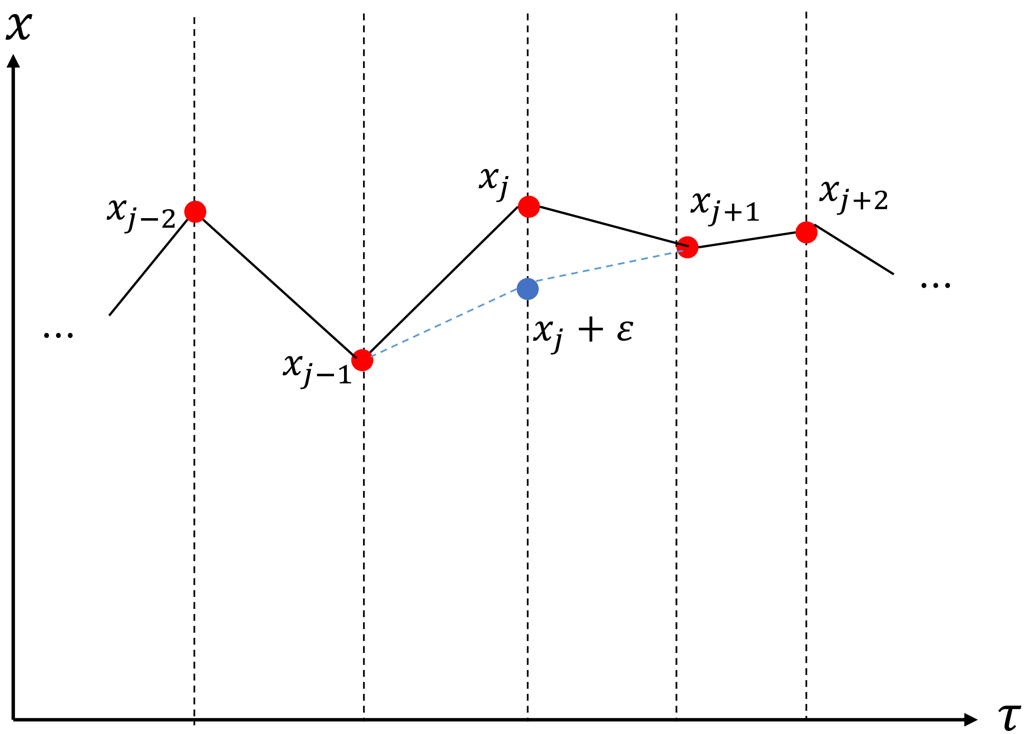
$$Z = \int \mathcal{D}x\ e^{-S_E[x(\tau)]/\hbar}, \quad x(0) = x(\hbar\beta)\ (\tau = \hbar\beta)$$

Quantum mechanics → Statistical physics

$$P_n = \frac{1}{Z} e^{-S_E[x(\tau)]/\hbar}, \quad \langle f(x,p) \rangle = \frac{1}{Z} \int \mathcal{D}x\ f(x,p) e^{-S_E[x(\tau)]/\hbar}, \cdots$$

PIMC

- Quantum Monte Carlo



- ➢ Metropolis algorithm $x_j \to x_j + \varepsilon$
- ➢ ε: random number
- ➢ The same method as the Ising model

$$\text{Acceptance probability} = \begin{cases} 1, & \Delta S_E < 0 \\ \exp(-S_E/\hbar), & \Delta S_E > 0 \end{cases}$$

$$S_E[x(\tau)] = \int_0^{\hbar\beta} d\tau \left(\frac{1}{2}m\left(\frac{dx}{d\tau}\right)^2 + V(q(\tau))\right)$$
$$\approx h\sum_i \left(\frac{1}{2}m\left(\frac{x_{j+1}-x_j}{h}\right)^2 + V(x_i)\right)$$
$$(h = \tau_{j+1} - \tau_j)$$

PIMC

- What we can get?

1. Ground state probability density

$$\psi(x) = \sum_{n=0}^{\infty} c_n e^{-iE_n t/\hbar} \psi_n(x) = \sum_{n=0}^{\infty} c_n e^{-E_n \tau/\hbar} \psi_0(x)$$

$$\approx e^{-E_0 \tau/\hbar} \psi_0(x) \quad (\tau \to \infty)$$

2. Expectation value of physical quantity

$$\langle f(x,p) \rangle$$

3. Correlation function

$$G(\tau) = \langle x(\tau) x(0) \rangle \sim e^{-(E_1 - E_0)\tau}$$

4. Energy gap

$$E_1 - E_0 = \frac{1}{h} \ln \frac{G(\tau)}{G(\tau + h)}$$

E_0: Ground state energy
E_1: First excited state energy

PIMC: Harmonic oscillator

1. Harmonic oscillator

- Potential

$$V(x) = \frac{1}{2} m\omega^2 x^2$$

- Exact solutions ($m = \hbar = 1$)

$$\psi_0(x) = \left(\frac{\omega}{\pi}\right)^{1/4} e^{-\omega x^2/2}, \quad |\psi_0(x)|^2 = \left(\frac{\omega}{\pi}\right)^{1/2} e^{-\omega x^2}$$

$$E_n = \omega\left(n + \frac{1}{2}\right), \ (n = 0, 1, 2, \cdots)$$

- Expectation value

$$\langle x^2 \rangle = \frac{1}{2\omega}$$

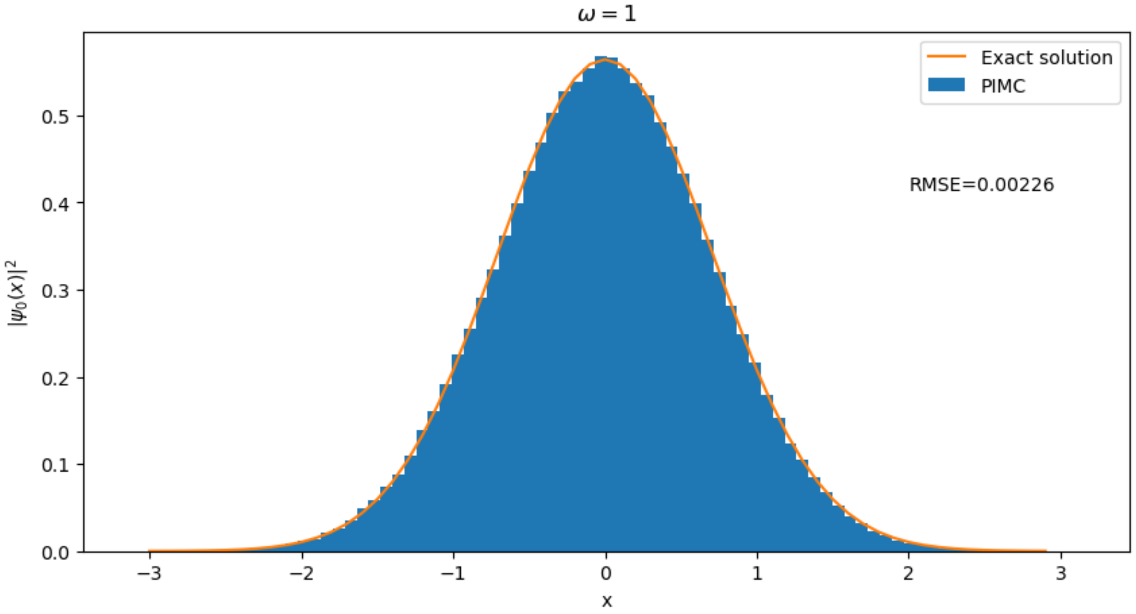
- Correlation function

$$G(\tau) \sim e^{-\omega\tau}, \quad E_1 - E_0 = \omega = \frac{1}{h} \ln \frac{G(\tau)}{G(\tau + h)}$$

PIMC: Harmonic oscillator

- Probability density: $\omega = 1$

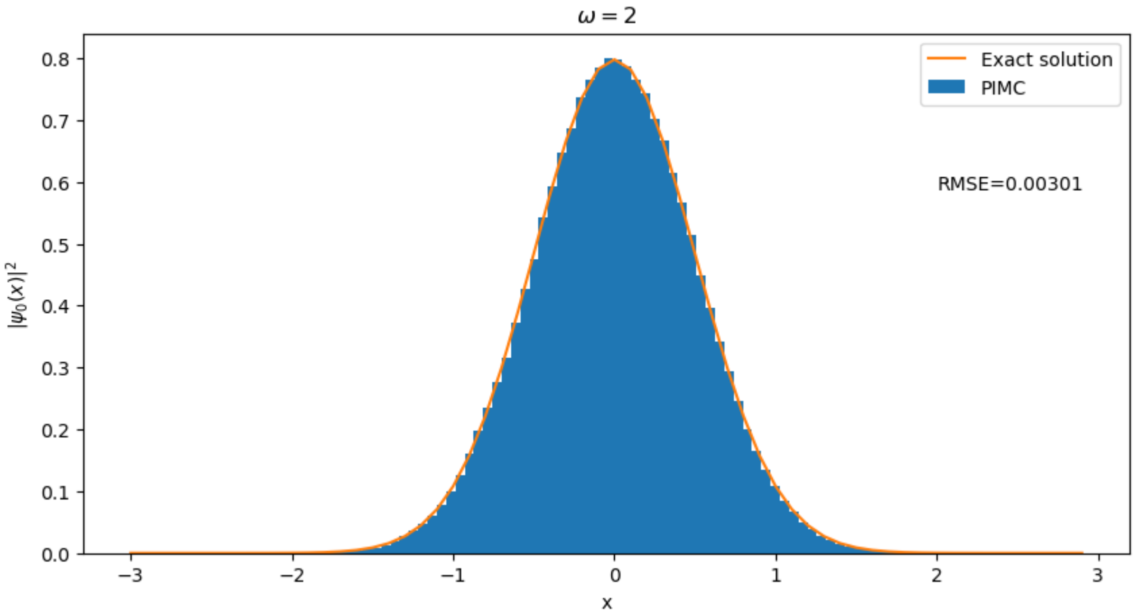
$$|\psi_0(x)|^2 = \left(\frac{\omega}{\pi}\right)^{1/2} e^{-\omega x^2}$$



PIMC: Harmonic oscillator

- Probability density: $\omega = 2$

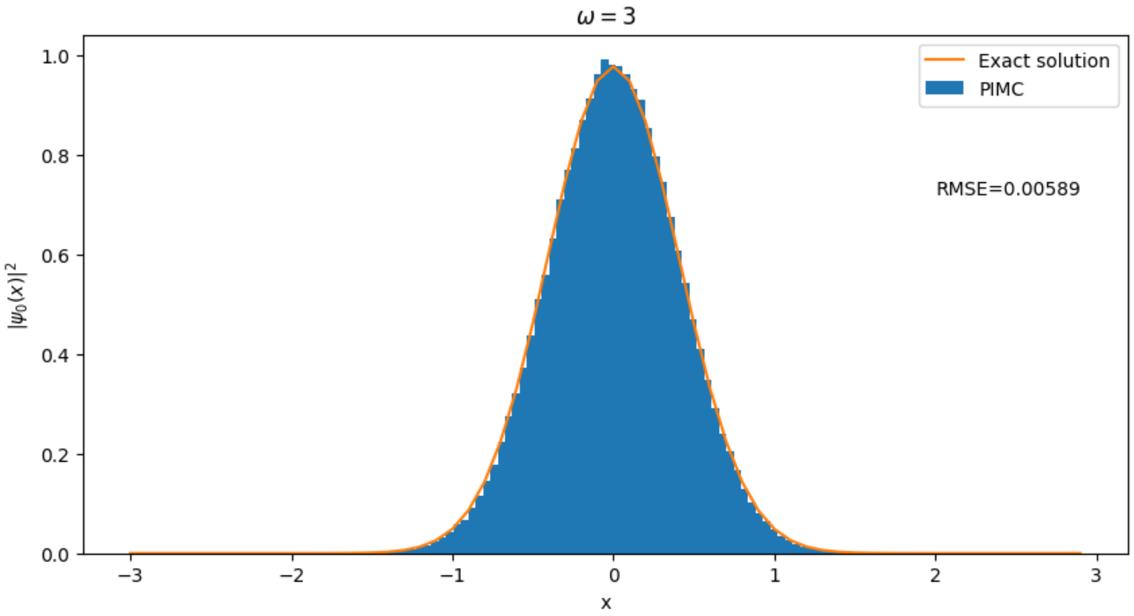
$$|\psi_0(x)|^2 = \left(\frac{\omega}{\pi}\right)^{1/2} e^{-\omega x^2}$$



PIMC: Harmonic oscillator

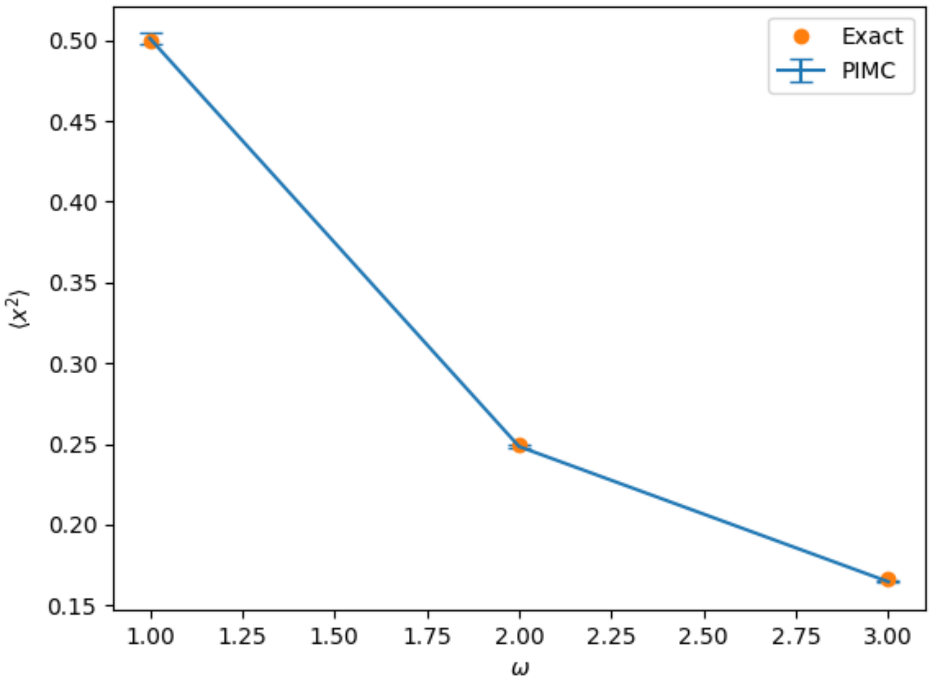
- Probability density: $\omega = 3$

$$|\psi_0(x)|^2 = \left(\frac{\omega}{\pi}\right)^{1/2} e^{-\omega x^2}$$



PIMC: Harmonic oscillator

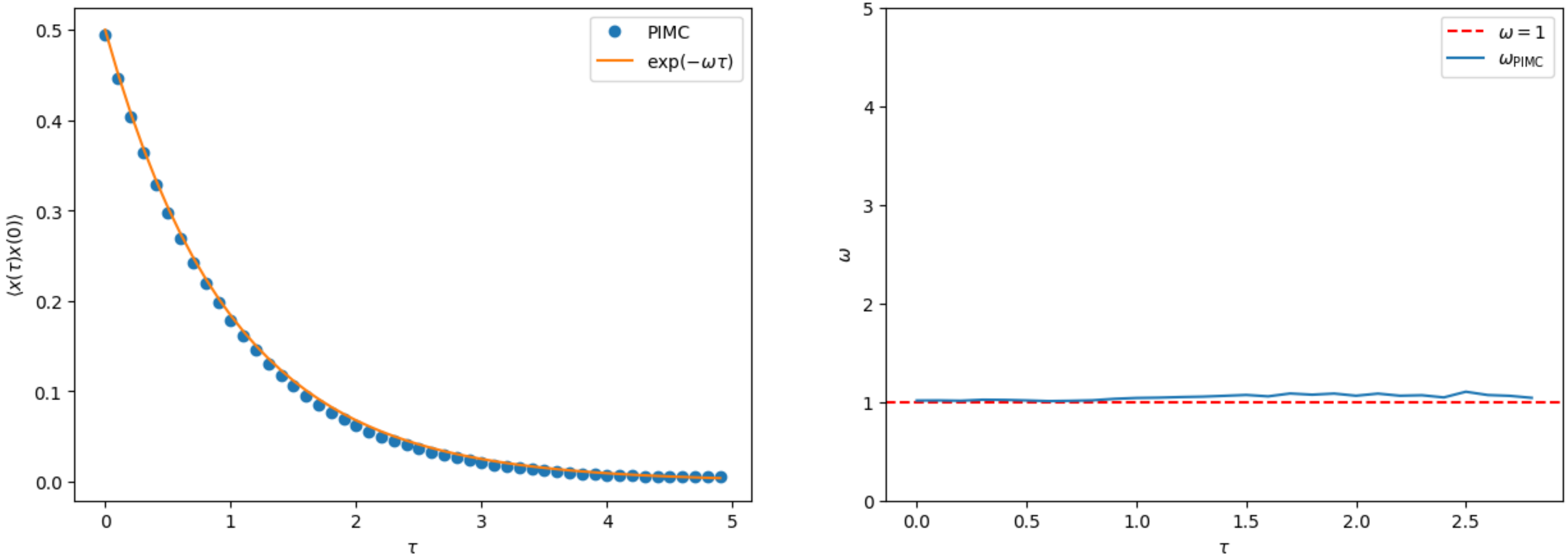
- $\langle x^2 \rangle$ comparison



ω	$\langle x^2 \rangle_{exact}$	$\langle x^2 \rangle_{PIMC}$
1	0.5	0.501 ± 0.0032
2	0.25	0.248 ± 0.0011
3	0.16666	0.165 ± 0.0006

PIMC: Harmonic oscillator

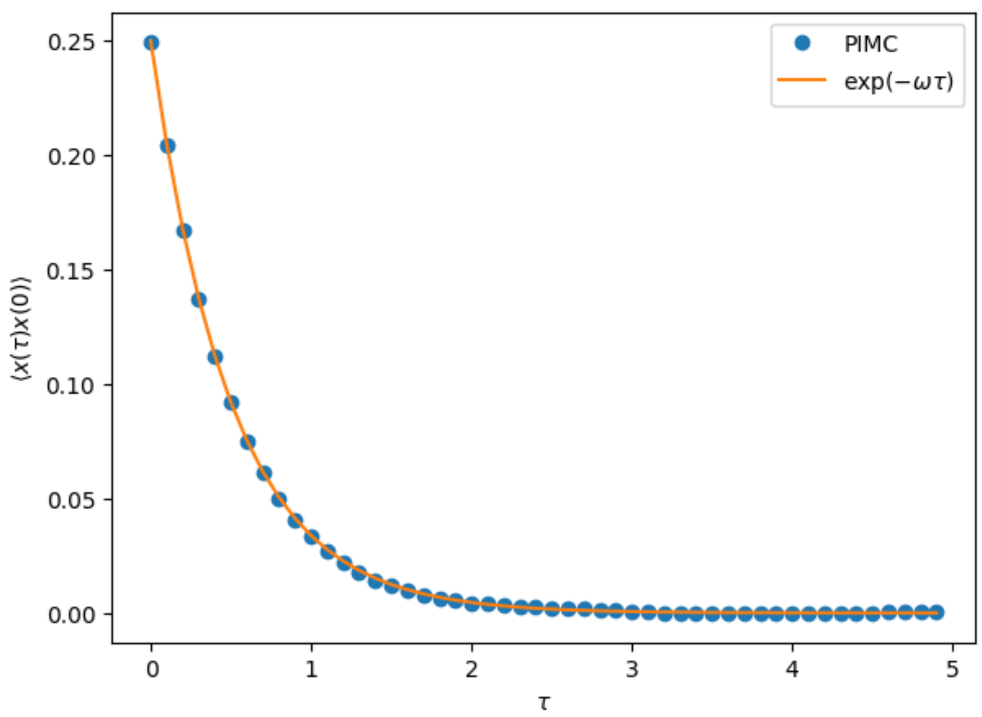
- Correlation function and energy gap: $\omega = 1$

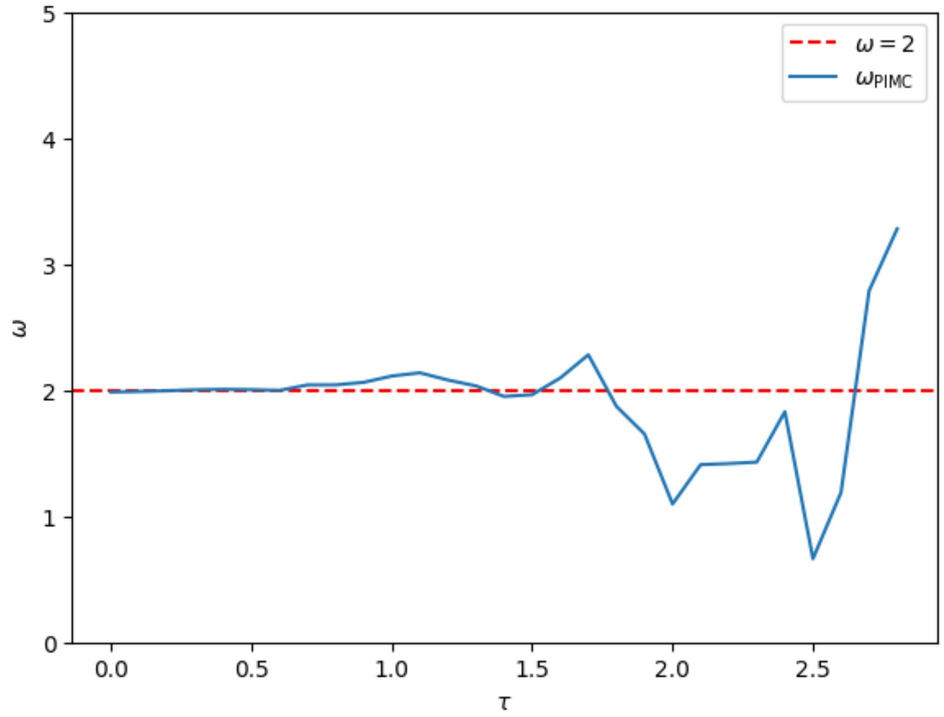


$$E_1 - E_0 = \frac{1}{h} \ln \frac{G(\tau)}{G(\tau + h)}$$

PIMC: Harmonic oscillator

- Correlation function and energy gap: $\omega = 2$

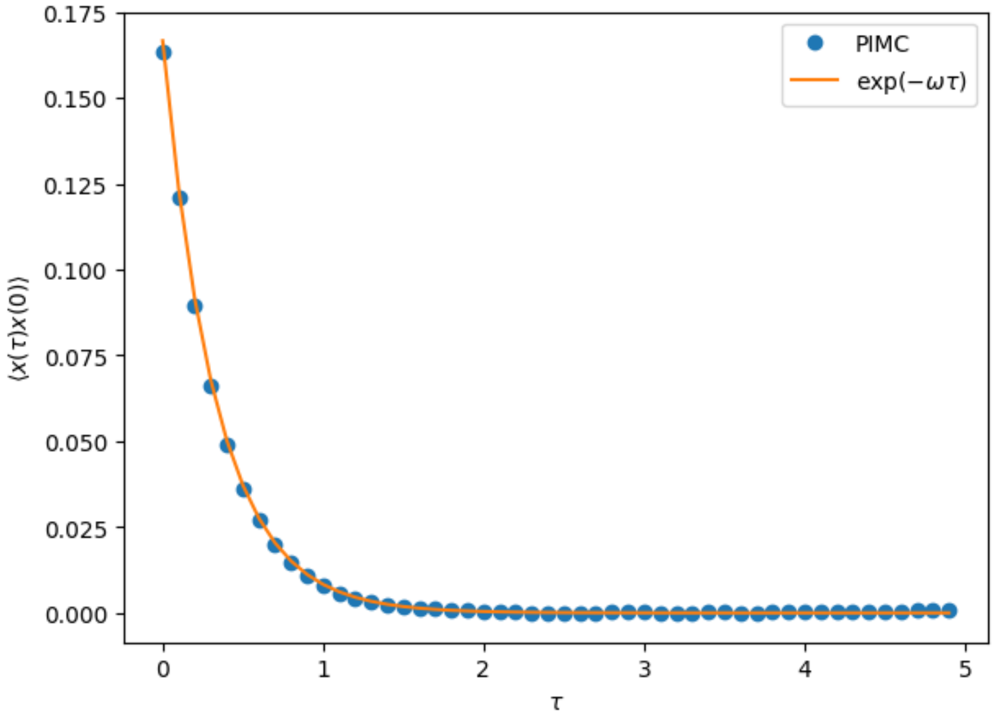


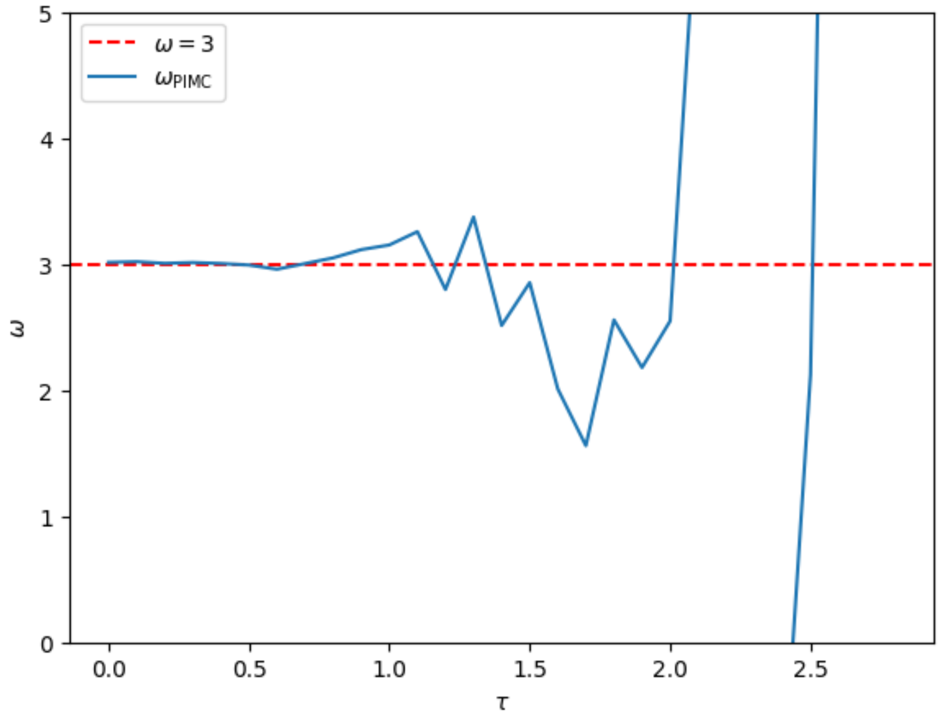


$$E_1 - E_0 = \frac{1}{h} \ln \frac{G(\tau)}{G(\tau + h)}$$

PIMC: Harmonic oscillator

- Correlation function and energy gap: $\omega = 3$





$$E_1 - E_0 = \frac{1}{h} \ln \frac{G(\tau)}{G(\tau + h)}$$

PIMC: Anharmonic oscillator

1. Anharmonic oscillator

- Potential

$$V(x) = V_0(x^2 - a^2)^2$$

- Exact solutions $(m = \hbar = 1)$

No exact solution

- Expectation value

No exact solution

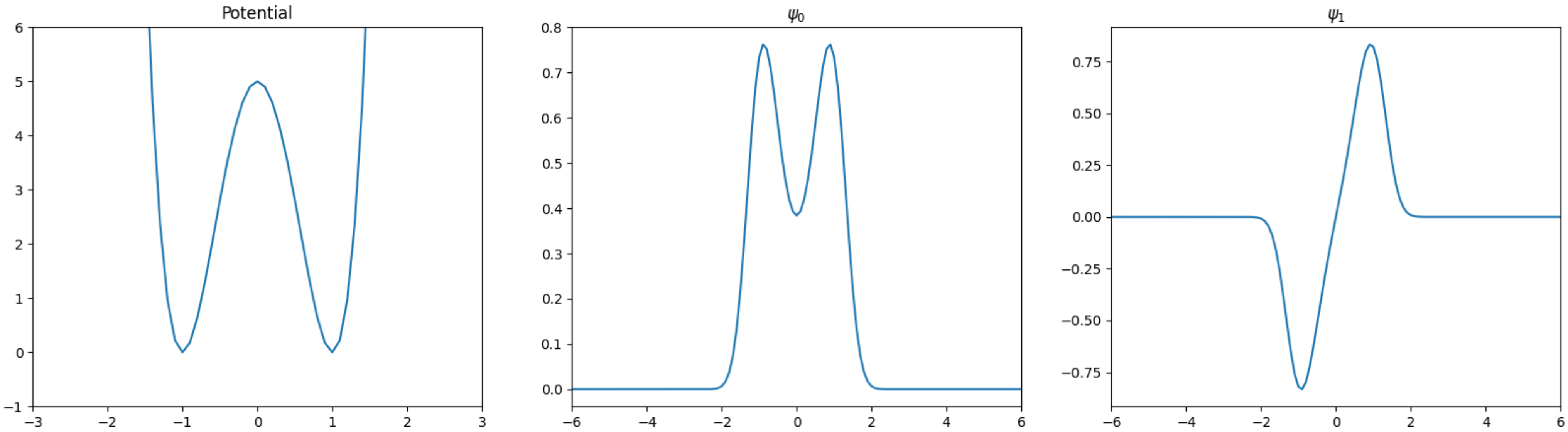
- Correlation function

No exact solution

PIMC: Anharmonic oscillator

1. Anharmonic oscillator

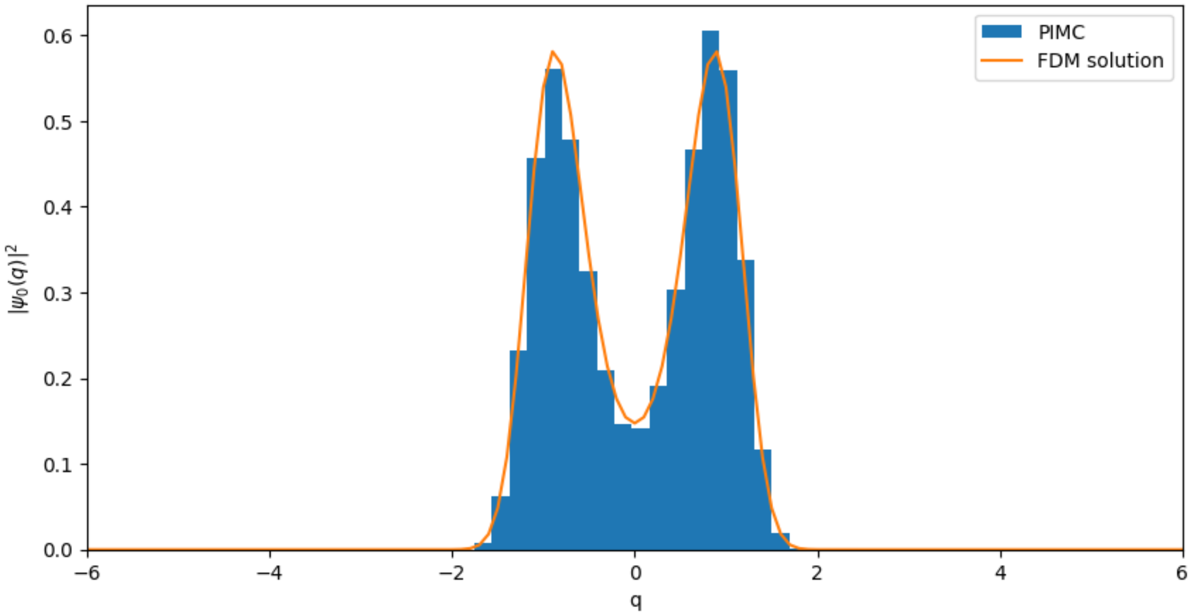
- Finite difference method $V(x) = V_0(x^2 - a^2)^2, \quad (V_0 = 5,\ a = 1)$



$$E_0 = 2.6583,\ \ E_1 = 3.0264$$
$$\Delta E = 0.3681$$

PIMC: Anharmonic oscillator

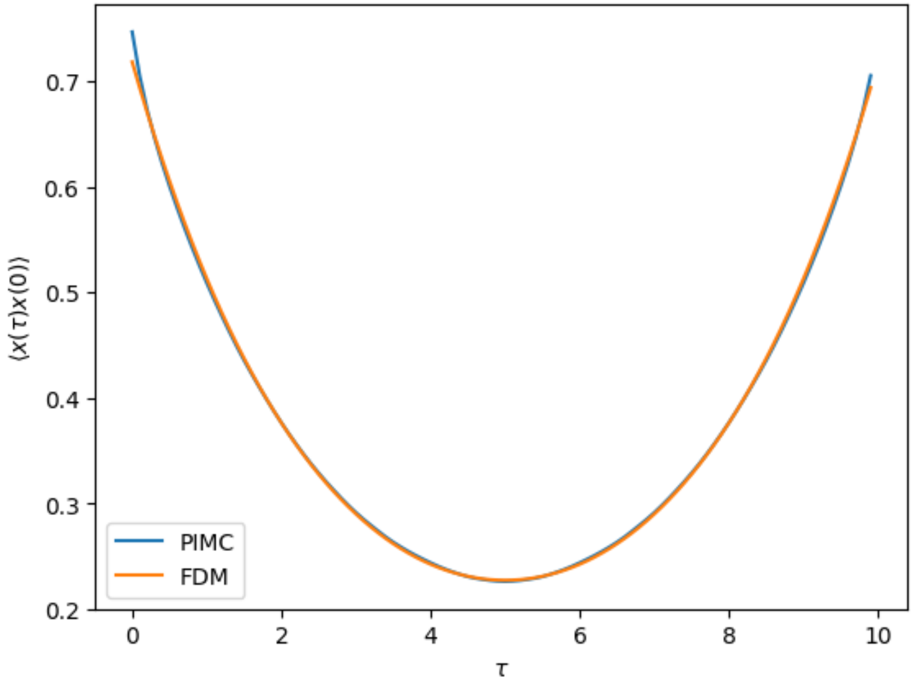
1. Anharmonic oscillator



- $\langle x^2 \rangle_{\mathrm{FDM}} = 0.7328$
- $\langle x^2 \rangle_{\mathrm{PIMC}} = 0.7467 \pm 0.0016$
- Relative error= 1.9%

PIMC: Anharmonic oscillator

1. Anharmonic oscillator



- $\Delta E_{\mathrm{FDM}} = 0.3681$
- $\Delta E_{\mathrm{PIMC}} = 0.3638 \pm 0.0007$
- Relative error= 1.2%

Is this efficient?

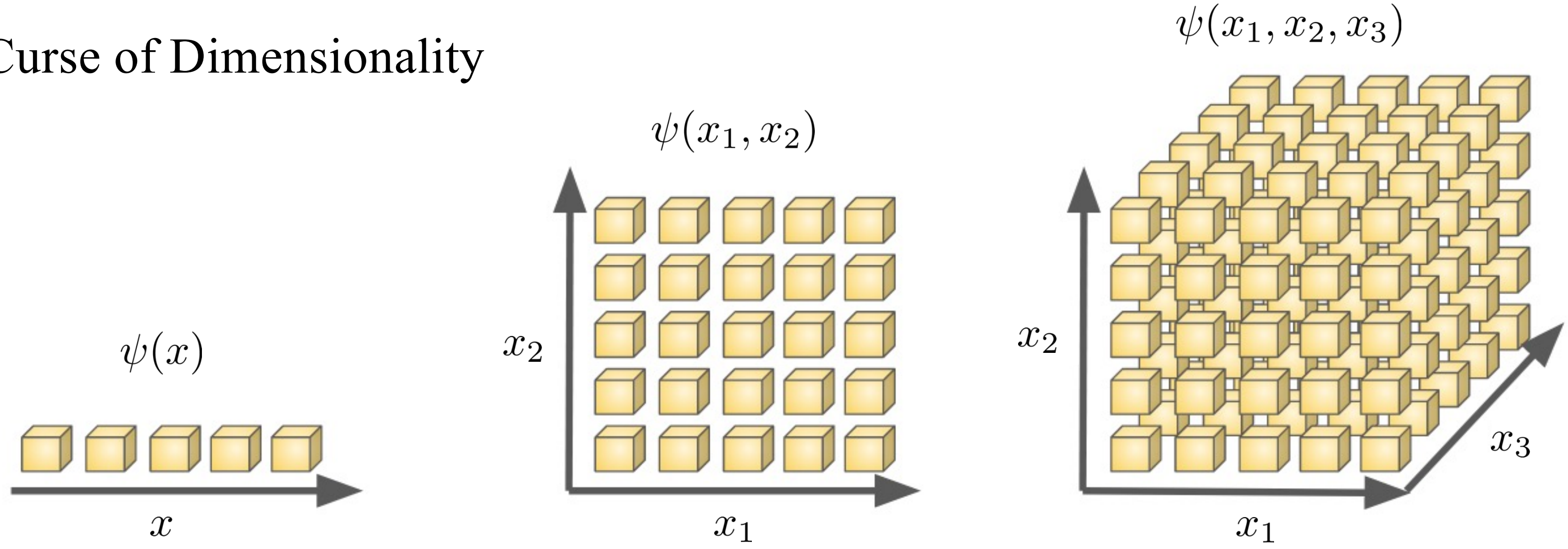
No

- Harmonic oscillator: We know the exact solution
- Anharmonic oscillator: Finite difference method is much faster

PIMC

Why we need PIMC?

- Curse of Dimensionality



- FDM: Explicit $\psi(x_1, x_2, \cdots, x_N)$
- PIMD: Path sampling

PIMC for Many-body Problem

- Permutation sampling case

PIMC for Many-body Problem

- Permutation sampling: 2 identical particles

Classical (Distinguishable) particles: $\Psi(x_1, x_2) = \psi_a(x_1)\psi_b(x_2) \neq \psi_a(x_2)\psi_b(x_1) = \Psi(x_2, x_1)$

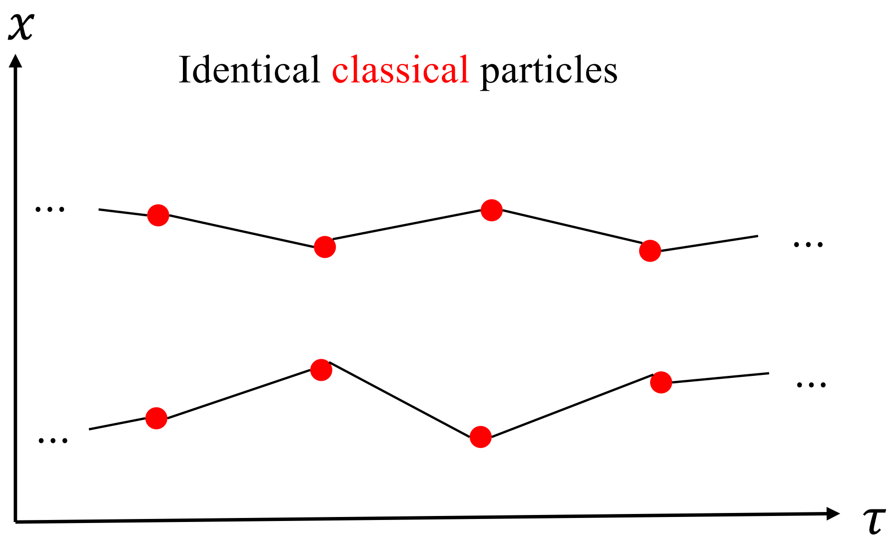
Quantum particles:

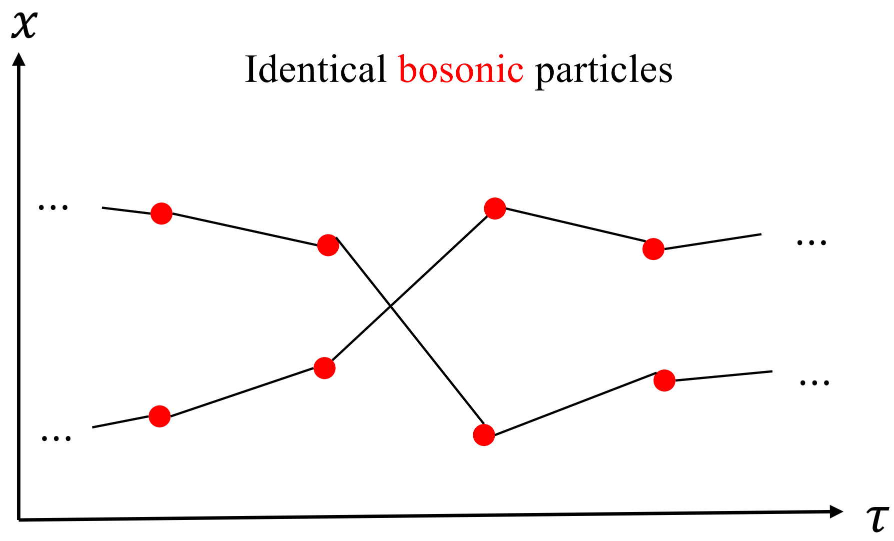
$$\begin{cases} \text{Boson:} & \Psi(x_1, x_2) = \dfrac{\psi_a(x_1)\psi_b(x_2) + \psi_b(x_1)\psi_a(x_2)}{\sqrt{2}} = \Psi(x_2, x_1) \\[2ex] \text{Fermion:} & \Psi(x_1, x_2) = \dfrac{\psi_a(x_1)\psi_b(x_2) - \psi_b(x_1)\psi_a(x_2)}{\sqrt{2}} = -\Psi(x_2, x_1) \end{cases}$$

➢ Indistinguishability of quantum particles

PIMC for Many-body Problem

- Permutation sampling: 2 identical particles

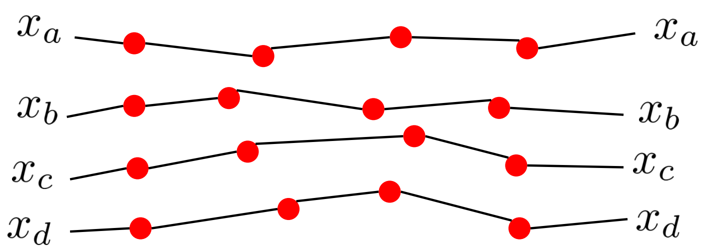




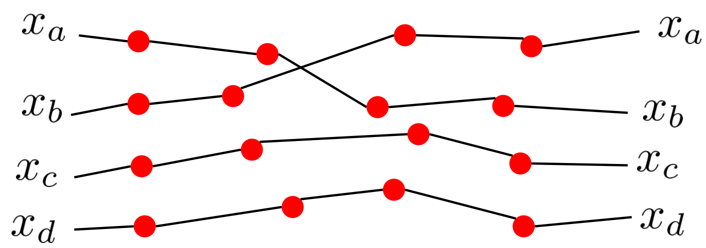
PIMC for Many-body Problem

- Permutation sampling: 4 identical particles

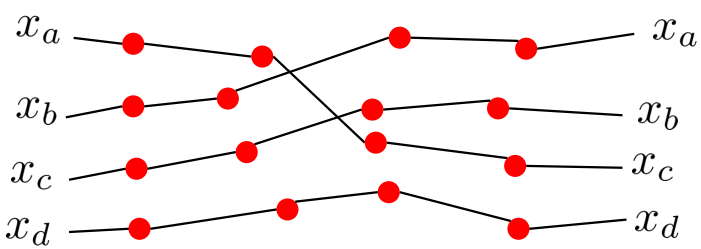
$$x(0) = x(\hbar\beta)\ (\tau = \hbar\beta)$$



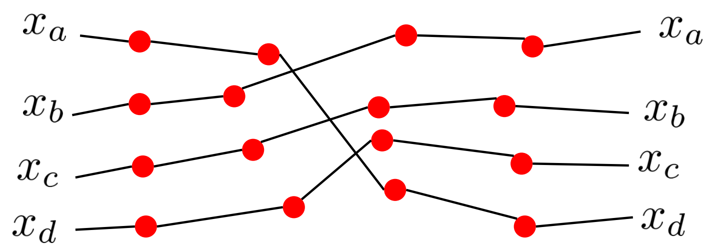
Cycle length = (1,1,1,1)



Cycle length = (2,1,1)



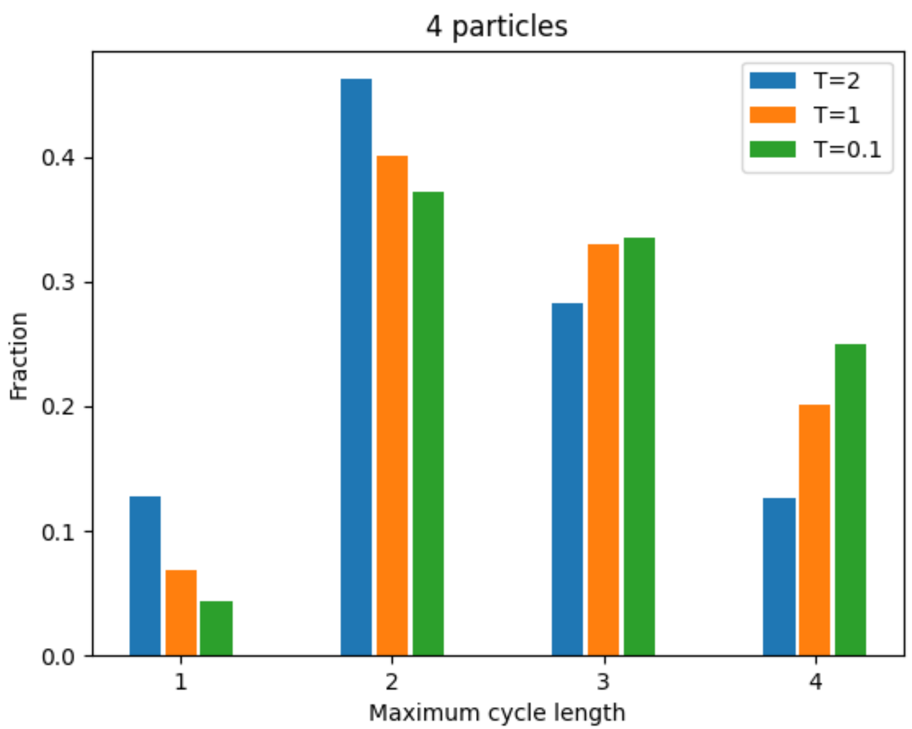
Cycle length = (3,1)



Cycle length = (4)

PIMC for Many-body Problem

- Permutation sampling: 4 identical particles



- ➢ Not only random displacement, but also particle exchange proposal
- ➢ Particle behaves as classical particle in high temperature ($\lambda_T \propto T^{-1/2}$)
- ➢ Lower temperature enhances probability of longer permutation cycles
- ➢ Growing importance of bosonic exchange statistics.

PIMC

- PIMC can be used for
  - Quantum mechanics (Many-body)
  - Field theory
  - Polymer physics
  - Financial markets

- PIMC can be implemented
  - Metropolis algorithm
  - Bisection algorithm
  - Worm algorithm

Thank you